

DATA SCIENCE CAPSTONE · PART 1

Lightning Talk #1

Three simple project ideas — problem, goals, data, and first look

 Credit Card Fraud

 Cardiovascular Risk

 Flight Delay

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What I will show today



01

Credit Card Fraud Detection

Find fake card payments among 1.85M+ transactions.

Features **23 cols**

Task type **Binary class.**



02

Cardiovascular Disease Prediction

Predict heart disease from 70K patient records.

Features **13 cols**

Task type **Binary class.**



03

Flight Delay Prediction

Predict delay minutes for 7M+ U.S. flights in 2024.

Features **35 cols**

Task type **Regression**



TOPIC 01

Credit Card Fraud Detection

PROBLEM STATEMENT

Credit card fraud costs banks a lot of money and puts customers at risk. Finding fake payments quickly helps keep people and banks safe. This project studies payment data to build a model that spots fraud.

GOALS

- ◆ Look at the data and find fraud patterns
- ◆ Study the data closely (EDA)
- ◆ Build and test a few models
- ◆ Pick the best model for finding fraud

DATA SNAPSHOT

SOURCE

Kaggle — Credit Card Fraud Detection Dataset

ROWS

1,852,394

COLUMNS

23 features

TARGET

is_fraud (0 = legit, 1 = fraud)

TASK

Binary Classification



Credit Card Fraud Detection

PROBLEM TYPE

Classification

target is_fraud is binary (0/1)

MISSING VALUES

0

no fixing needed

FEATURE MIX

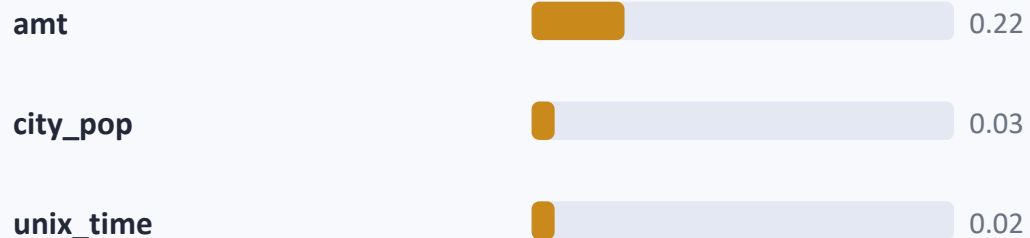
12 cat / 10 num

categorical vs. numerical

MITIGATION PLAN

99.42% real vs. 0.58% fraud — a big imbalance. Plan: use class weighting, SMOTE, and precision/recall scores instead of accuracy.

STRONGEST CORRELATIONS WITH TARGET



OUTLIERS

amt and city_pop have extreme values

Location fields (lat/long) also have outliers. Large amounts and big cities are normal, so we will keep and check these values instead of removing them.



TOPIC 02

Cardiovascular Disease Prediction

PROBLEM STATEMENT

Heart disease is one of the top causes of death in the world. Predicting it early can help doctors find high-risk patients and treat them sooner. This project studies patient health data to predict heart disease.

GOALS

- ◆ Look at patient health data and risk factors
- ◆ Study the data closely (EDA)
- ◆ Build models to predict heart disease
- ◆ Test the models and pick the best one

DATA SNAPSHOT

SOURCE

Kaggle — Cardiovascular Disease Dataset

ROWS

70,000

COLUMNS

13 features

TARGET

cardio (0 = no disease, 1 = disease)

TASK

Binary Classification



Cardiovascular Disease Prediction

PROBLEM TYPE

Classification

target cardio is binary and well balanced

MISSING VALUES

0

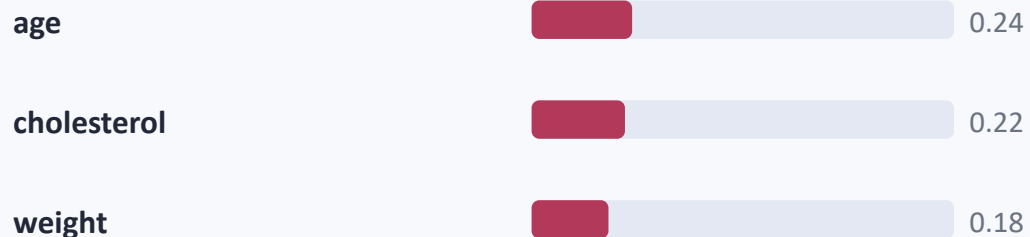
no fixing needed

FEATURE MIX

6 cat / 5 num

categorical vs. numerical

STRONGEST CORRELATIONS WITH TARGET



OUTLIERS

ap_hi and ap_lo have unrealistic values

Height and weight have some extreme but possible values. Blood pressure (ap_hi and ap_lo) likely has typing errors that need fixing before we build the model.



TOPIC 03

Flight Delay Prediction

PROBLEM STATEMENT

Flight delays affect millions of passengers every year. They cost money and make customers unhappy. Predicting delays before takeoff can help airlines plan better and give passengers better information.

GOALS

- ✦ Look at past flight data
- ✦ Study the data closely (EDA)
- ✦ Build new features carefully, without data leakage
- ✦ Build a model to predict arrival delay in minutes

DATA SNAPSHOT

SOURCE

Kaggle — Flight Delay Dataset (2024)

ROWS

7M+

COLUMNS

35 features

TARGET

arr_delay (minutes, continuous)

TASK

Regression



Flight Delay Prediction

PROBLEM TYPE

Regression

target arr_delay is a number with a long tail

MISSING VALUES

Several cols

mostly from cancelled flights

FEATURE MIX

9 cat / 26 num

categorical vs. numerical

STRONGEST CORRELATIONS WITH TARGET



⚠ We found possible data leakage — some time fields will be checked and fixed before training the model.

OUTLIERS

Delay and distance values have a long tail

arr_delay, dep_delay, taxi time, air time, and distance all have outliers. This is normal for real flights. We will keep and check these values, instead of removing them, to tell real long delays apart from data errors.